

Evaluation

Tierce Exams - 50%

Assignments - 15%

Project - 20%

Lab Exam - 10%

Attendance - 5%

Attendance is mandatory.

Syllabus

Basic Text Processing: Tokenization, Stemming, Spelling Correction;

Language Modeling: N-grams, smoothing;

Syntax: PCFGs, Dependency Parsing;

Semantics: Word Sense Disambiguation, Word2Vec, GloVE

Syllabus

Deep Learning for NLP: Basic and Advanced Models (Seq-to-Seq, BERT), Parts of Speech Tagging, Transformer;

LLMs - Instruction Tuning, Prompt Tuning, Alignment, RAGs

Applications: Information Extraction: Relation extraction; Named entity recognition, coreference resolution, Machine Translation;

Resources

Books -

- Daniel Jurafsky and James H. Martin. 2021. Speech and Language Processing. 3rd Edition, <https://web.stanford.edu/~jurafsky/slp3/>
- Delip Rao, Brian McMahan, Natural Language Processing with PyTorch: Build Intelligent Language Applications Using Deep Learning. O'Reilly Publishers
- Hobson Lane, Cole Howard, Hannes Hapke. 2019. Natural Language Processing in Action. Link to Live Book - <https://livebook.manning.com/book/natural-language-processing-in-action/about-this-book/>

Resources

Course content will be adapted from -

- Slides of the books if available online.
- Course content of popular NLP courses offered in institutes like Stanford, IITD, IITKGP etc.
- Material available for online courses.

Proper Acknowledgement will be provided for every case.